The many slopes of the Phillips curve: evidence from U.S. industries

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Introduction

- **Question.** How does cost pass-through differ across industries?
  - Sectoral pass-throughs central to predicting inflation and to optimal policy.
- **Literature.** Papers using cross-sectional identification estimate a common slope.
- **This paper.** Construct marginal cost series for each U.S. industry and estimate heterogeneous pass-through to prices.
- **Findings.**
  - Fast industries adjust even faster than Pastén et al. (2020)-Calvo.
  - Slow industries adjust even slower than Pastén et al. (2020)-Calvo.
  - DC inflation lower overall but rose more in Covid.
  - Under corrected optimal policy, oil shocks generate less inflation.

Contents

1. Motivation

2. Data

3. Estimating equation, identification, and the shocks

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5. Implications

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Motivation

Why do we care? Sectoral Phillips curves

- Calvo pricing in industry i with rigidity θ_i and discount factor β:

$$p_{it} = \theta_i p_{i,t-1} + (1-\theta_i)(1-\beta\theta_i)\sum_{k=0}^{\infty}(\beta\theta_i)^k \mathbb{E}_t \mathrm{mc}_{i,t+k}$$

$$\implies \pi_{it} = \kappa_i\left(\mathrm{mc}_{it} - p_{it}\right) + \beta\,\mathbb{E}_t\pi_{i,t+1}, \quad \kappa_i \equiv \frac{(1-\theta_i)(1-\beta\theta_i)}{\theta_i}$$

- Marginal cost in industry i:

$$\mathrm{mc}_{it} = \Omega_i\,\boldsymbol{p}_t + \Omega_{iL}\,w_{it} - a_{it}$$

where Ω is the input-output matrix and a_{it} is TFP.

$$\text{Stacking} \implies \boldsymbol{\pi}_t = \mathrm{diag}(\boldsymbol{\kappa})(\boldsymbol{mc}_t - \boldsymbol{p}_t) + \beta\mathbb{E}_t\boldsymbol{\pi}_{t+1}.$$

Why do we care? Predicting inflation

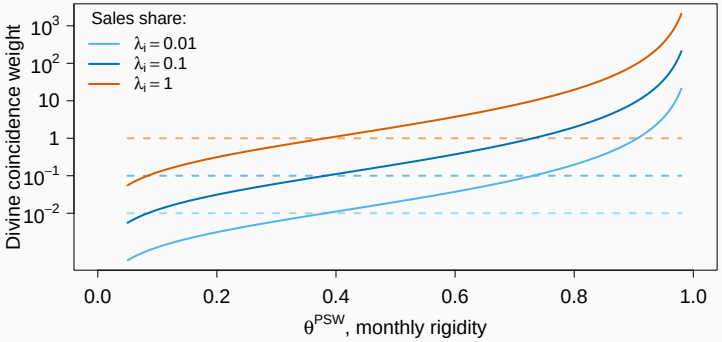
- A scalar inflation measure, such as the CPI or the PCE, is defined by a weight vector $\boldsymbol{b}$.

$$\pi_t^b = \boldsymbol{b}' \operatorname{diag}(\boldsymbol{\kappa})(\boldsymbol{mc}_t - \boldsymbol{p}_t) + \beta \mathbb{E}_t \pi_{t+1}^b.$$

- The forecast depends on the industry slopes weighted by those weights, $b_i \kappa_i$.

Why do we care? Optimal policy

- Rubbo (2023): Divine coincidence index weights industry i by λ_i/κ_i.



- Pass-through speed is at least as important as the sales share.

Literature

- **Time series estimates of the Phillips curve slope.** Galí and Gertler (1999), Mavroeidis et al. (2014).
- **Cross-sectional estimates.** Gagliardone et al. (2025), Hazell et al. (2022), Herreño et al. (2026).
  - Estimate a common κ across units.
  - Valid only under homogeneous passthrough; cannot span key positive and normative implications.
  - Pass-through by adjustment frequency and through the network: Gopinath and Itskhoki (2010), Luo and Villar (2022), Minton and Wheaton (2023).
- **Price rigidity and cost passthrough.** Use measured adjustment frequencies. Pastén et al. (2020), Carvalho et al. (2021), Bils and Klenow (2004), Nakamura and Steinsson (2008), Carvalho (2006), Nakamura and Steinsson (2010), Auclert et al. (2024), Alvarez et al. (2016).
  - Do not directly estimate passthrough.
- **Networks, aggregation, and policy.** La'O and Tahbaz-Salehi (2022), Rubbo (2023).

Data

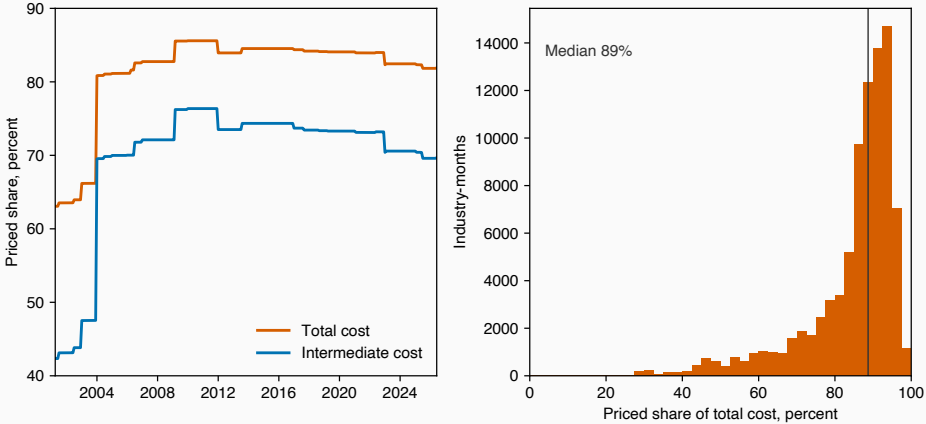
Prices and the measured input-cost series

- **Prices.** The industry's published PPI, 286 BEA industries.
- **Cost.** The IO-weighted sum of domestic input prices plus ECI wages,

$$\Delta \widetilde{\mathrm{mc}}_{it} = \sum_j \omega_{ij}^{v} \, \Delta p_{jt} + \omega_{iL}^{v} \, \Delta w_{it}^{ECI},$$

- We take ω from the input-output tables at vintage $v \in \{2007, 2012, 2017\}$.
- **Issues.**
  - **Coverage.** Some supplier prices may be missing. We do not renormalize.
  - **TFP.** The TFP component of marginal costs is unobserved, hence the tilde.

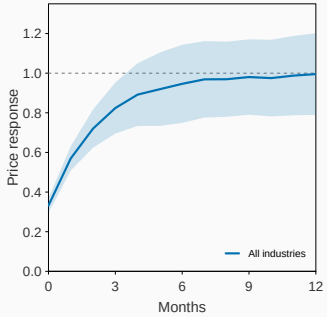
Coverage



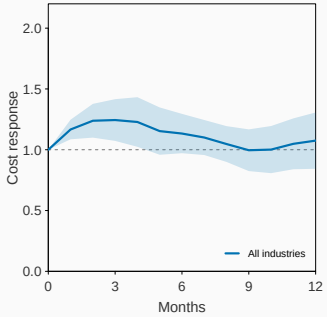
Note: Left panel shows Domar-weighted share of legend-denoted cost that has a measured PPI. Right panel shows the same share over every industry-month.

Reduced-form pass-through

$$\Delta_h p_{i,t+h} = \beta_h \Delta \mathrm{mc}_{i,t} + \gamma X_{it} + e_{it}$$



$$\Delta_h \mathrm{mc}_{i,t+h} = c_h \Delta \mathrm{mc}_{i,t} + \gamma X_{it} + e_{it}$$



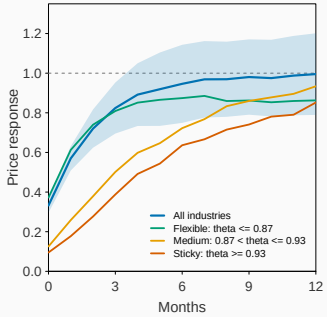
A permanent nominal marginal cost shock implies full long-run pass-through.

X_{it}: 12 lags of $\Delta \mathrm{mc}$ and of Δp. Industry and time fixed effects. Driscoll–Kraay SEs (Bartlett, bandwidth $\max(12, h+1)$). The cost series used here leaves out the own-sector price component.

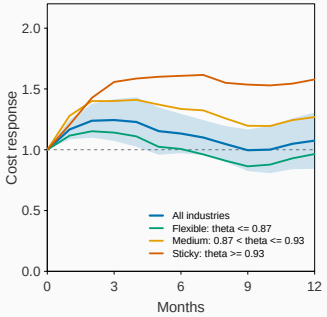
▸ Coverage split ▸ 2 years

Reduced-form pass-through

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$$\Delta_h \mathrm{mc}_{i,t+h} = c_h \Delta \mathrm{mc}_{i,t} + \gamma X_{it} + e_{it}$$



High θ industries pass through slowest.

X_{it}: 12 lags of $\Delta \mathrm{mc}$ and of Δp. Industry and time fixed effects. Driscoll–Kraay SEs (Bartlett, bandwidth $\max(12, h+1)$). The cost series used here leaves out the own-sector price component.

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Estimating equation, identification, and the shocks

Passthrough in the Calvo model

- Standard Calvo pricing model for each industry i,

$$p_{it} = \theta_i p_{i,t-1} + (1-\theta_i)(1-\beta\theta_i) \sum_{k=0}^{\infty} (\beta\theta_i)^k \, \mathbb{E}_t \mathrm{mc}_{i,t+k}.$$

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Proposition: Price growth and cost growth

$$\Delta p_{s,t+H} = \sum_{j=0}^{\infty} w_{s,H,j}\,\Delta \mathrm{mc}_{s,t+j} + \nu_{s,t,H},$$

where

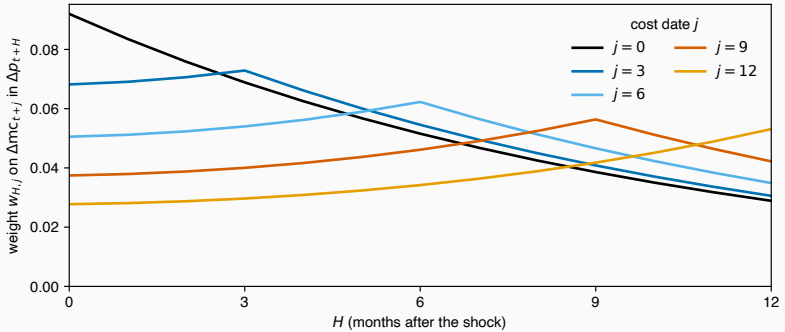
$$w_{s,H,j} = \frac{(1-\theta_s)\theta_s^H}{1-\beta\theta_s^2}\Big\{(1-\theta_s)(\beta\theta_s)^{j+1} + (1-\beta\theta_s)(\beta\theta_s^2)^{\max\{j-H,0\}}\theta_s^{-j}\Big\}.$$

and where the residual is orthogonal to any period-t news,

$$\mathbb{E}[\nu_{s,t,H}\varepsilon_t] = 0.$$

Passthrough in the Calvo model

- These $w_{s,H,j}$ weights are the **Jacobians** from news at 0 of a cost change at j into price growth at H.



$\theta = 0.908$, the median of the measured θ_s, and $\beta = 0.99^{1/3}$, the monthly discount factor.

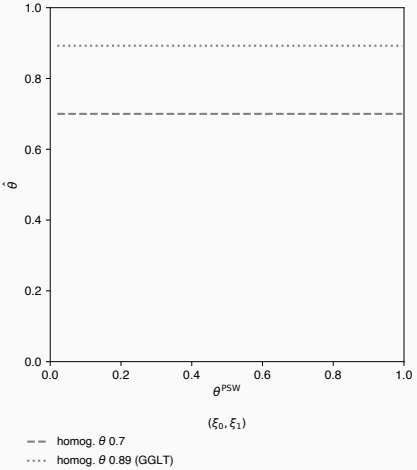
The power rule for the sectoral slopes

- We estimate a transformed Calvo slope:

$$\log \kappa_s = \xi_0 + \xi_1 \log \kappa_s^{\mathrm{PSW}}.$$

- $\kappa_s^{\mathrm{PSW}} \equiv \frac{(1-\theta_s)(1-\beta\theta_s)}{\theta_s}$ at the *measured* θ_s.

hypothesis	(ξ_0, ξ_1)
homogeneous pass-through	$(\log \kappa,\ 0)$



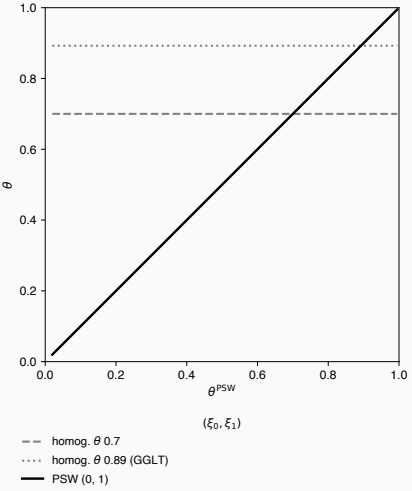
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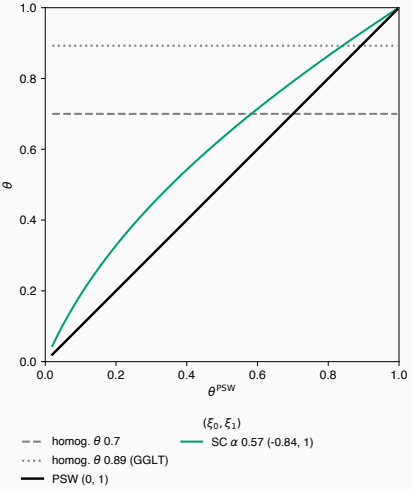
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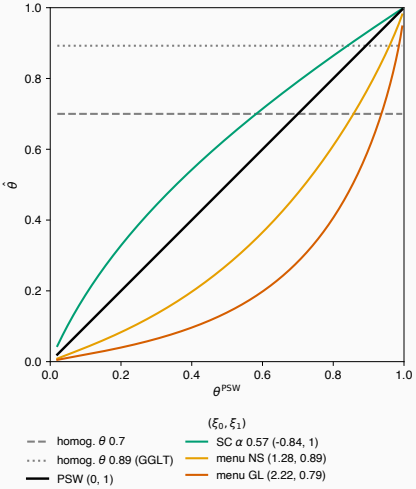
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strategic complementarity α	$(\log(1-\alpha),\ 1)$
menu costs (ARRS), constant Kur	$(\approx \log(6/\mathrm{Kur}),\ \approx 1)$

ARRS = Auclert et al. (2024): Calvo duration $\times\, \mathrm{Kur}/6$, Calvo has $\mathrm{Kur} = 6$.



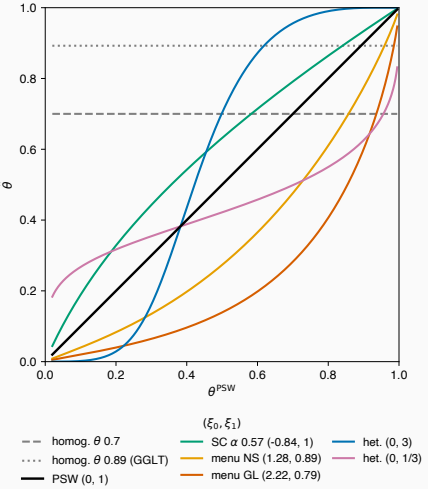
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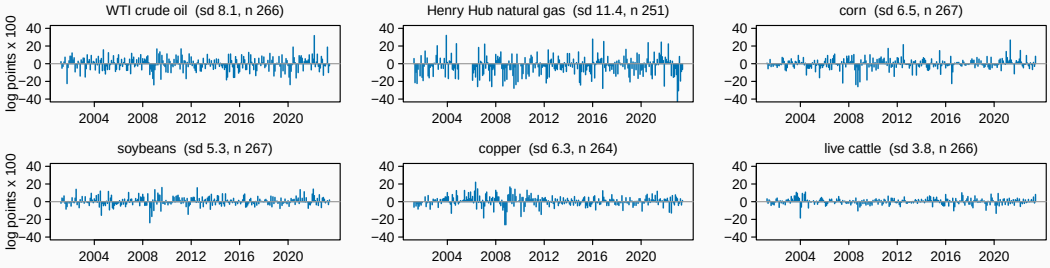
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menu costs (ARRS), constant Kur	$(\approx \log(6/\mathrm{Kur}),\ \approx 1)$
heterogeneity beyond f and Kur	$\xi_1 \neq 1$



Instrument conditions

- We need shocks that are **relevant** for nominal marginal costs and satisfy two **exclusion restrictions**:
  1. Occur exactly at time t, i.e. between PPI collection dates t and $t-1$
  2. Orthogonal to the sector's *own* TFP and markup shocks
- We'll use commodity futures surprises
- Use daily futures series to compute monthly shocks satisfying (1)
- Orthogonality to TFP and markup shocks in (2) comes from:
  - For each commodity shock, exclude the domestic commodity production sector in case that sector's TFP or markup shock loads on the global price movement
  - Control for aggregate TFP and markups with time FE
  - Looser restriction than in network pass-through estimation

The shocks: commodity futures surprises



- **Timing.** PPI prices are collected once a month, the Tuesday of the week of the 13th. ε_t is the price change, between the collection days of $t-1$ and t, of the nearest contract delivering after month t.
- **Why not spot price changes?** The spot change might have been forecastable (e.g. expected mean reversion).
- **Why not identified news shocks?** Not obviously uncorrelated with US markups and TFP, much less powered (relevant).

Optimal instruments

- A conditional moment restriction for every s, H and t: $\mathbb{E}[\nu_{s,t,H}(\boldsymbol{\xi}) \mid \varepsilon_t] = 0$.
- Chamberlain (1987): the efficient instrument for ξ_0 (similarly for ξ_1),

$$Z^{\xi_0}_{s,t,H} = \frac{\mathbb{E}[\partial \nu_{s,t,H}/\partial \xi_0 \mid \varepsilon_t]}{\mathrm{Var}(\nu_{s,t,H} \mid \varepsilon_t)} = -\frac{\sum_j \left(\partial w_{s,H,j}(\boldsymbol{\xi})/\partial \xi_0\right) \mathbb{E}[\Delta \mathrm{mc}_{s,t+j} \mid \varepsilon_t]}{\mathrm{Var}(\nu_{s,t,H} \mid \varepsilon_t)}.$$

  - We assume $\mathrm{Var}(\nu_{s,t,H} \mid \varepsilon_t)$ is constant.
  - The $w_{s,H,j}$ derivative is analytical.
  - We measure the conditional expectation of cost growth from a dynamic network model using PSW rigidity
- Summing over s, H and t, get **two instruments** for **two parameters**:

$$\sum_{s,H,t} Z^{\xi_0}_{s,t,H}\, \nu_{s,t,H}(\hat{\boldsymbol{\xi}}) = 0, \qquad \sum_{s,H,t} Z^{\xi_1}_{s,t,H}\, \nu_{s,t,H}(\hat{\boldsymbol{\xi}}) = 0.$$

▸ GMM weight ▸ Min. distance

Final Specification

$$\Delta p_{s,t+H} = \alpha_{t,H} + \sum_{j=0}^{36} w_{s,H,j}(\boldsymbol{\xi})\, \Delta \mathrm{mc}_{s,t+j} + e_{s,t,H}$$

Outcome	price growth at horizons $H = 0, \ldots, 11$ after the shock month t
$\alpha_{t,H}$	shock-month $\times$ horizon effect; absorbs the common response
$w_{s,H,j}(\boldsymbol{\xi})$	Calvo weights on the cost changes, with θ_s from the power rule
Costs	own-inclusive input price index; leads truncated at $j = 36$
Sample	286 industries, 267 shock months, 2001m4–2023m6
Shocks	six commodity futures surprises ε_t
Instruments	the two Chamberlain (1987) efficient instruments, constructed from theory alone (no many instruments issues)
Estimator	$\hat{\boldsymbol{\xi}}$ solves $\sum_{s,H,t} \tilde{Z}^{\xi_0}_{s,t,H}\, \tilde{e}_{s,t,H}(\hat{\boldsymbol{\xi}}) = 0$ and $\sum_{s,H,t} \tilde{Z}^{\xi_1}_{s,t,H}\, \tilde{e}_{s,t,H}(\hat{\boldsymbol{\xi}}) = 0$, tildes denote FE demeaning. Just-identified nonlinear IV via Gauss–Newton.

Results

Regression estimates

Results

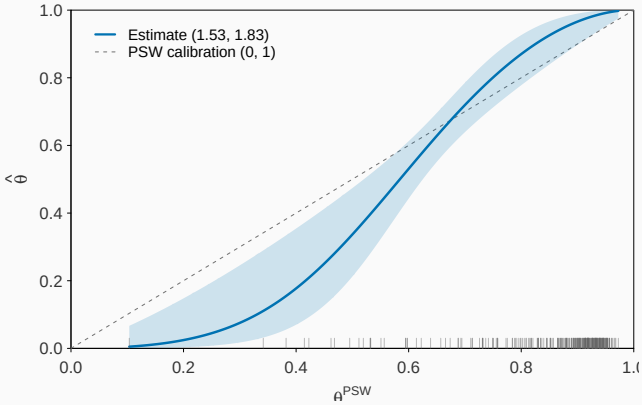
	Estimate of ξ	
	ξ_0	ξ_1
Estimate	1.53	1.83
Wald s.e.	(0.64)	(0.38)
AR 95% interval	[0.42, 3.31]	[1.19, 3.08]
AR p vs $(0, 1)$	0.025	
AR p vs $(\log \kappa, 0)$	<0.001	

First stage

	Dependent variable	
	$\partial X/\partial \xi_0$	$\partial X/\partial \xi_1$
Z^{ξ_0}	0.128^{***}	-0.030^{***}
	(0.013)	(0.010)
Z^{ξ_1}	-0.021	0.134^{***}
	(0.015)	(0.027)
SW conditional F	133.7	122.2
Rank test p-value	<0.001	

DK(12) standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. First-stage coefficients are standardized.

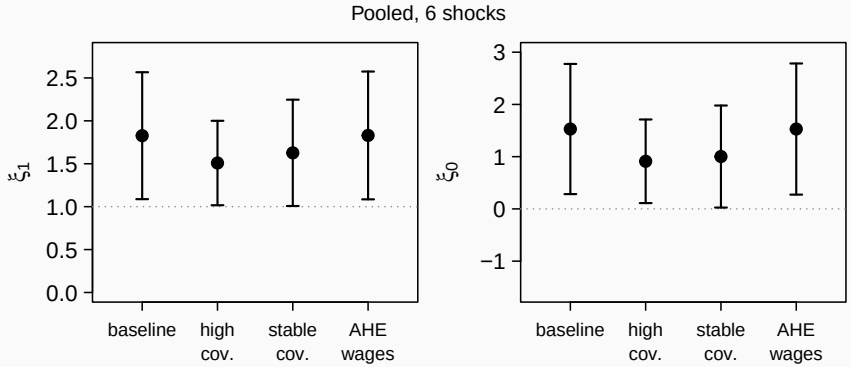
Industry passthrough in the data



The shaded region is a 95% DK(12) CI.

- Faster passthrough in fast industries. Slower passthrough in slow industries.

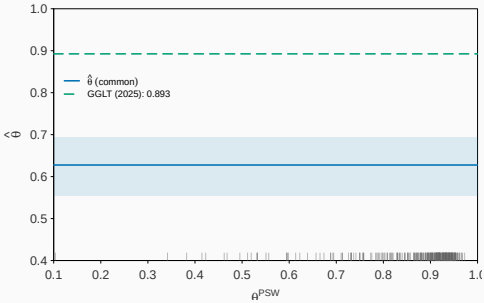
Robustness



95% Wald, DK(12) CIs (dotted lines at the PSW values $\xi_1 = 1$, $\xi_0 = 0$). High coverage: input cost coverage share of cost stays at or above 0.80 over the 36 cost months; Stable: the share moves at most 0.05 within the cost window; AHE wages replace the ECI term.

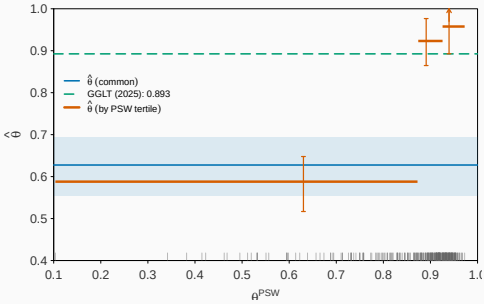
A homogeneous slope is a LATE

- Impose $\xi_1 = 0$ and estimate a common θ for every sector, compare to Gagliardone et al. (2025).



A homogeneous slope is a LATE

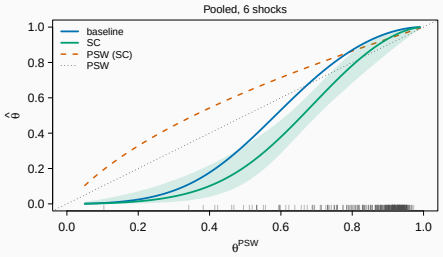
- Impose $\xi_1 = 0$ and estimate a common θ for every sector, compare to Gagliardone et al. (2025).
- Now do this by PSW tertile.



- GGLT's firms are sticky, or their instruments target sticky firms.

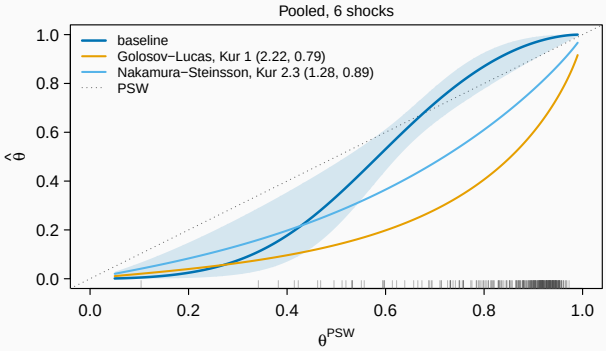
Strategic complementarity

- Strategic complementarities lead to altered passthrough: $\tilde{\kappa}_i = (1-\alpha)\,\kappa_i$.
- We re-estimate with the optimal reset price in X targeting $(1-\alpha)\,\mathrm{mc}_{it} + \alpha\, p_{it}$, the α-weighted average of marginal cost and own price; also incorporate SC into the instrument's network model.
- We use $\alpha = 0.570$ from Gagliardone et al. (2025).



- Even bigger discrepancy for low θ industries!

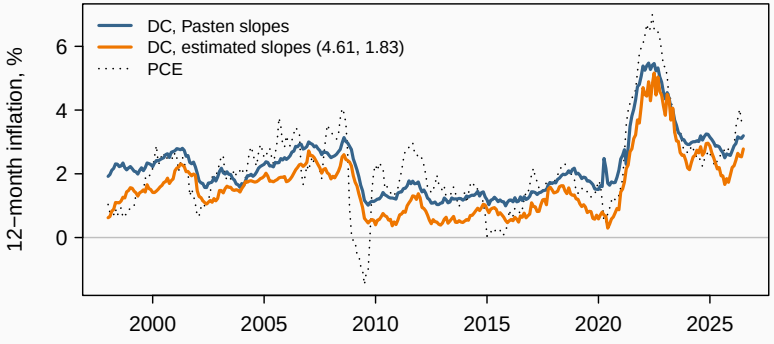
Menu costs



- Menu costs fix pass-through for flexible sectors at the expense of the sticky sectors.

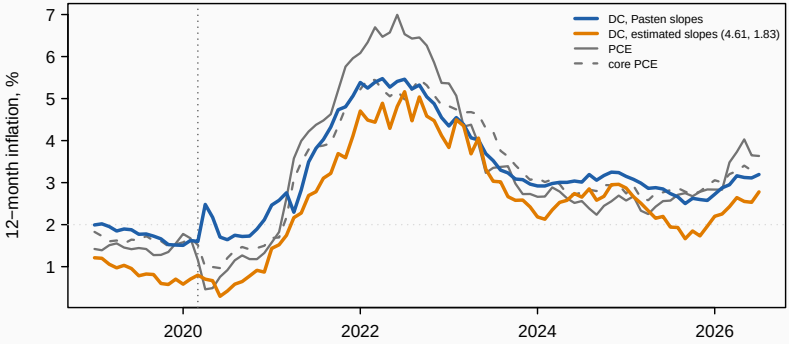
Implications

Divine coincidence inflation



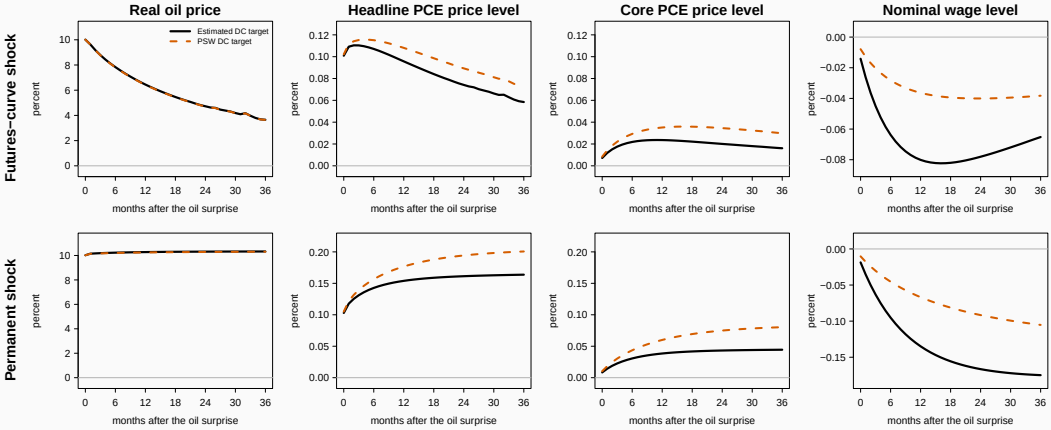
▸ Methodology ▸ Wages through the rule

Divine coincidence inflation



▸ Methodology ▸ Wages through the rule

An oil shock under two DC targets



Conclusion

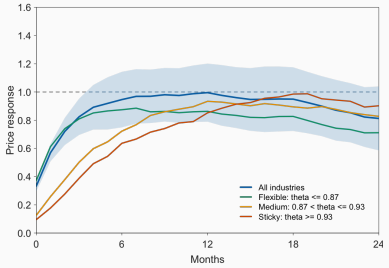
Conclusion

- Existing cross-sectional estimates of the NKPC slope are homogeneous
- We estimate heterogeneous slopes, finding steeper slopes than PSW for flexible sectors and flatter slopes than PSW for sticky sectors
- Baseline strategic complementarity and menu cost models cannot account for these facts
- DC inflation under our estimated slopes rose more during COVID and has climbed more sharply since late 2025
- Targeting our DC inflation measure entails less look-through for oil shocks

Backup

2 years

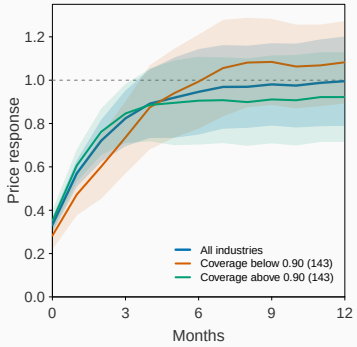
$$p_{s,t+h} - p_{s,t-1} = b_h \Delta \mathrm{mc}_{s,t} + \sum_{k=1}^{12} \gamma_k \Delta \mathrm{mc}_{s,t-k} + \sum_{k=1}^{12} \delta_k \Delta p_{s,t-k} + \alpha_i + \alpha_t + e_{st}$$



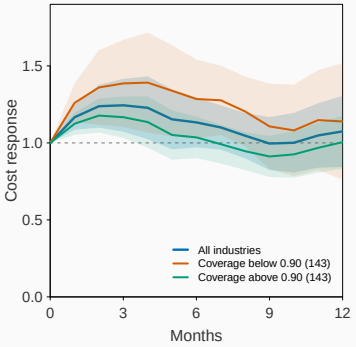
Back

Pass-through by coverage

$$\Delta_h p_{i,t+h} = \beta_h \Delta \mathrm{mc}_{i,t} + \gamma X_{it} + e_{it}$$



$$\Delta_h \mathrm{mc}_{i,t+h} = c_h \Delta \mathrm{mc}_{i,t} + \gamma X_{it} + e_{it}$$



Back

Backup: the same estimator as efficient GMM

- One unconditional moment per (s, H), $286 \times 12 = 3{,}432$ of them: $m_{s,H}(\boldsymbol{\xi}) = \mathbb{E}[\varepsilon_t \, \nu_{s,t,H}(\boldsymbol{\xi})] = 0$.
- Efficient GMM: $\min_{\boldsymbol{\xi}} \; \hat{\boldsymbol{m}}(\boldsymbol{\xi})' \, \boldsymbol{V}^{-1} \hat{\boldsymbol{m}}(\boldsymbol{\xi})$ with $\boldsymbol{V} = \mathrm{Var}(\hat{\boldsymbol{m}})$. First-order conditions, two equations:

  $$\boldsymbol{J}' \boldsymbol{V}^{-1} \hat{\boldsymbol{m}}(\hat{\boldsymbol{\xi}}) = \boldsymbol{0}, \qquad J^{\xi_0}_{s,H} = \frac{\partial m_{s,H}}{\partial \xi_0} = \mathbb{E}\left[\varepsilon_t \, \frac{\partial \nu_{s,t,H}}{\partial \xi_0} \right] = -\,\mathrm{Var}(\varepsilon_t) \sum_j \frac{\partial w_{s,H,j}}{\partial \xi_0} \, k_{s,j}, \quad J^{\xi_1}_{s,H} \text{ likewise.}$$

- With $\mathrm{Var}(\nu_{s,t,H} \mid \varepsilon_t)$ constant, $\boldsymbol{V} \propto \boldsymbol{I}$ and the combination $\boldsymbol{a}^* = \boldsymbol{V}^{-1} \boldsymbol{J}$ gives

  $$\sum_{s,H} J^{\xi_0}_{s,H} \, \hat{m}_{s,H}(\hat{\boldsymbol{\xi}}) \;\propto\; \sum_{s,H,t} Z^{\xi_0}_{s,t,H} \, \nu_{s,t,H}(\hat{\boldsymbol{\xi}}) = 0, \qquad \text{and the same for } \xi_1,$$

  the two optimal-instrument equations.
- Difference from Chamberlain (1987): GMM combines the *moments* $\mathbb{E}[\varepsilon_t \, \nu_{s,t,H}]$; Chamberlain conditions the *residuals* $\nu_{s,t,H}$ on ε_t and builds the instrument from $\mathbb{E}[\partial \nu_{s,t,H} / \partial \boldsymbol{\xi} \mid \varepsilon_t]$. They coincide here because the conditional mean of costs is linear in ε_t.

A nonlinear conditional mean would make Chamberlain's instrument nonlinear in ε_t, which no weighting of these moments reproduces. The composite row sets $\mathrm{Var}(\varepsilon_{k,t})$ constant across the six shocks.

◂ Back

Backup: the same estimator as minimum distance

- IRFs on the surprise for every (s, H) and (s, j) (time $\times$ horizon demeaned): $\hat{\pi}_{s,H}$ from $\Delta p_{s,t+H}$ on ε_t, $\hat{\gamma}_{s,j}$ from $\Delta \mathrm{mc}_{s,t+j}$ on ε_t.
- Because $\nu_{s,t,H} = \Delta p_{s,t+H} - \sum_j w_{s,H,j}\, \Delta \mathrm{mc}_{s,t+j}$, each (s, H) moment is a distance between the price IRF and the Calvo filter of the cost IRF:

$$\hat{m}_{s,H}(\boldsymbol{\xi}) = \widehat{\mathrm{Var}}(\varepsilon_t)\left[\hat{\pi}_{s,H} - \sum_j w_{s,H,j}(\boldsymbol{\xi})\, \hat{\gamma}_{s,j}\right].$$

- The two estimating equations are the normal equations of weighted minimum distance,

$$\sum_{s,H} J^{\xi_0}_{s,H}\left[\hat{\pi}_{s,H} - \sum_j w_{s,H,j}(\hat{\boldsymbol{\xi}})\, \hat{\gamma}_{s,j}\right] = 0, \qquad J^{\xi_0}_{s,H} \propto \sum_j \frac{\partial w_{s,H,j}}{\partial \xi_0}\, k_{s,j},$$

  and the same with $J^{\xi_1}_{s,H}$: the Jacobian's cost IRFs are taken from the model ($k_{s,j}$) rather than estimated ($\hat{\gamma}_{s,j}$), so nothing is fitted in sample.
- Reading: choose the rule under which Calvo-filtered relative cost IRFs match relative price IRFs.

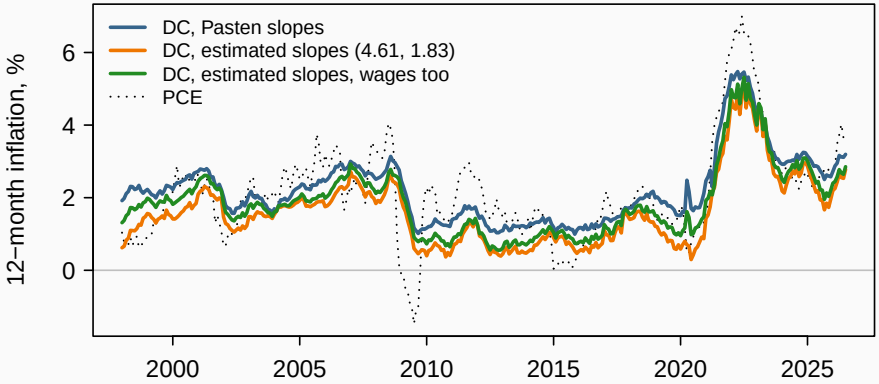
◂ Back

Backup: how the DC index is built

- **Rubbo's recipe.** Every BEA detail sector weighted by λ_i/κ_i, Domar weight over Calvo slope (her $\lambda_i(1-\hat{\delta}_i)/\hat{\delta}_i$ is the same object). Wages are a sector: average hourly earnings, quarterly adjustment probability 0.25. Own PPI first; a sector without one takes the PCE components it sells to, at producers' values. Weights renormalised each month over the sectors observed.
- **What we change.** 401 sectors of the 2017 detail, Domar weights of the IO vintage in force (her 405 of 2012, fixed). PPIs as published, not seasonally adjusted, hence 12-month rates. Pasten's backfilled frequencies where the micro data have none (she used the mean). No back-extension: 1997m1 to 2026m7, services from 2006. Covered Domar weight 0.76 before 2006, 0.90 after.
- **Held fixed across the two series.** All of the above. Only the slope changes: (a) the Calvo κ_i at the PSW frequency; (b) $\kappa_i = e^{\xi_0}(\kappa_i^{\mathrm{PSW}})^{\xi_1}$ at (1.53, 1.83). The wage slope is the Calvo one in both: wages are 26% of the index under (a), 2% under (b).

▸ Wages through the rule ◂ Back

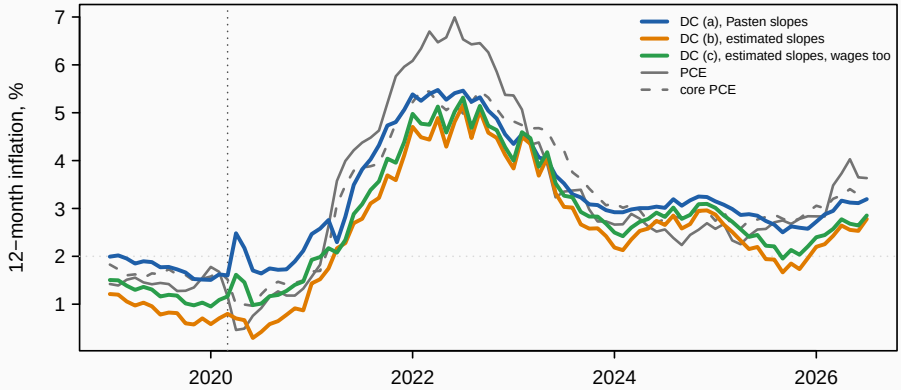
Backup: wages through the power rule too



Applying the rule to the wage slope as well (wage θ from 0.91 to 0.97) lifts the wage weight from 2% to 15% of the index. The series rises by 0.2 points and keeps its shape: the same peak month, the same return below 3% in September 2023.

◂ Back

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◂ Back

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